

Functional Test Cases: Authentication & Authorization

Library Management System

Test Design Team (Non-Coders/Specification Experts)

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Note: These test cases are designed from a black-box, user-centric perspective. Actual results and status are to be filled during test execution. Refer to the SRS for complete requirements.

I.A Functional Test Cases: Authentication & Authorization

Table 1: Authentication and Authorization Test Cases

Test Case ID	Test Title	Test Steps	Expected Results
Sub-Section: User Registration (by Librarian)			
TC_AUTH_REG_001	Successful new user (Member) registration by Librarian	- Login as Librarian. - Navigate to 'User Management' or 'Add User' section. - Fill in all required fields with valid and unique data for a new 'Member' (e.g., username, password, email, full name). - Submit the registration form.	- System accepts the data. - A success message "User registered successfully" (or similar) is displayed. - The new user appears in the list of users. - The new user can subsequently log in (to be tested in Login cases).
TC_AUTH_REG_002	Attempt to register a new user with an existing username	- Login as Librarian. - Navigate to 'Add User' section. - Fill in user details, using a username that already exists in the system. - Submit the registration form.	- System rejects the registration. - An appropriate error message is displayed (e.g., "Username already exists"). - The user is not created.
TC_AUTH_REG_003	Attempt to register a new user with missing required fields	- Login as Librarian. - Navigate to 'Add User' section. - Fill in user details but leave one or more mandatory fields blank (e.g., password or email). - Submit the registration form.	- System rejects the registration. - An appropriate error message is displayed, indicating which fields are required. - The user is not created.
TC_AUTH_REG_004	Attempt to register a new user with invalid data format (e.g., email)	- Login as Librarian. - Navigate to 'Add User' section. - Fill in user details, providing an incorrectly formatted email address (e.g., "usertest.com"). - Submit the registration form.	- System rejects the registration. - An appropriate error message is displayed (e.g., "Invalid email format"). - The user is not created.
Sub-Section: User Login			
TC_AUTH_LOGIN_001	Successful login with valid Librarian credentials	- Navigate to the login page. - Enter valid Librarian username. - Enter corresponding valid password. - Click the 'Login' button.	- User is authenticated successfully. - User is redirected to the Librarian dashboard/homepage. - Librarian-specific functionalities are visible/accessible.

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Table 1 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_AUTH_LOGIN_002	Successful login with valid Member credentials	• Navigate to the login page. • Enter valid Member username. • Enter corresponding valid password. • Click the 'Login' button.	• User is authenticated successfully. • User is redirected to the Member dashboard/homepage. • Member-specific functionalities are visible/accessible.
TC_AUTH_LOGIN_003	Attempt login with invalid username	• Navigate to the login page. • Enter a username that does not exist. • Enter any password. • Click the 'Login' button.	• Login fails. • An appropriate error message is displayed (e.g., "Invalid username or password"). • User remains on the login page.
TC_AUTH_LOGIN_004	Attempt login with valid username but invalid password	• Navigate to the login page. • Enter a valid existing username. • Enter an incorrect password for that username. • Click the 'Login' button.	• Login fails. • An appropriate error message is displayed (e.g., "Invalid username or password"). • User remains on the login page.
TC_AUTH_LOGIN_005	Attempt login with empty username field	• Navigate to the login page. • Leave the username field empty. • Enter any password. • Click the 'Login' button.	• Login fails. • An appropriate error message is displayed (e.g., "Username is required"). • User remains on the login page.
TC_AUTH_LOGIN_006	Attempt login with empty password field	• Navigate to the login page. • Enter a valid username. • Leave the password field empty. • Click the 'Login' button.	• Login fails. • An appropriate error message is displayed (e.g., "Password is required"). • User remains on the login page.
Sub-Section: User Logout			
TC_AUTH_LOGOUT_001	Successful logout for a logged-in Librarian	• Login as Librarian. • Click the 'Logout' button/link.	• User is logged out successfully. • User is redirected to the login page or public homepage. • Session is terminated (user cannot access protected pages without logging in again).
TC_AUTH_LOGOUT_002	Successful logout for a logged-in Member	• Login as Member. • Click the 'Logout' button/link.	• User is logged out successfully. • User is redirected to the login page or public homepage. • Session is terminated.
Sub-Section: Role-Based Access Control (RBAC)			

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Test Case ID	Test Title	Test Steps	Expected Results
TC_AUTH_RBAC_001	Verify Librarian can access Librarian-specific features (e.g., User Management)	• Login as Librarian. • Attempt to navigate to 'User Management' section. • Attempt to navigate to 'Book Management' (add/edit/delete books).	• Librarian successfully accesses 'User Management' page. • Librarian can view and interact with user administration tools. • Librarian successfully accesses 'Book Management' page and can perform CRUD operations on books.
TC_AUTH_RBAC_002	Verify Member cannot access Librarian-specific features (e.g., User Management)	• Login as Member. • Check if 'User Management' link/button is visible. • Attempt to navigate directly to the User Management URL (if known). • Check if 'Add Book' or 'Edit Book' options are available.	• 'User Management' link/button is not visible or is disabled. • Direct navigation to User Management URL redirects to an error page or member dashboard. • 'Add Book'/'Edit Book' options are not available. Access is denied.
TC_AUTH_RBAC_003	Verify Member can access Member-specific features (e.g., Search Books, View Borrowal History)	• Login as Member. • Attempt to navigate to 'Search Books' page. • Attempt to view 'My Borrowal History'.	• Member successfully accesses 'Search Books' page and can search. • Member successfully accesses 'My Borrowal History' and can view their own history.
TC_AUTH_RBAC_004	Verify guest (not logged in) user access to public pages and restriction from protected pages	• Ensure no user is logged in (clear session/cookies or use incognito window). • Attempt to navigate to the login page. • Attempt to navigate to a public book search page (if applicable). • Attempt to navigate to 'User Management' URL. • Attempt to navigate to 'My Borrowal History' URL.	• Login page is accessible. • Public book search page is accessible (if such a feature exists for guests). • Access to 'User Management' URL is denied, likely redirecting to login page. • Access to 'My Borrowal History' URL is denied, likely redirecting to login page.